Team Breliconi – Animal Lift Project Elements

Element A-

Introduction:

Team Breliconi is our group of four students from Brentwood High School working on an engineering project for the Nashville Zoo. Our goal is to design and build a large animal lift. This lift will help the zoo's veterinary team lift, move and rehabilitate animals that weigh up to 2,000 pounds. This is a big challenge, but it's one we are excited to take on.

The team includes Ellison Abromowitz, Britton West, Corbin Wolfgang, and Nicolas Blucher. We are working with mentors Mike Kiser and Dr. Stacy Klein-Gardner, as well as zoo professionals Stephen Wancea and Dr. Heather Schwartz. Mr. Wancea is the Hospital Manager at the Nashville Zoo, and Dr. Schwartz is the Director of Animal Health. These people are helping guide us so we can solve a real problem in the zoo.

This project has been going on for three years. Each year, a new group of students takes over and improves the design. We are building off what past teams have done and trying to fix the biggest problem they faced: stability.

Identifying the Problem:

The main issue with the current lift design is stability. The lift is supposed to lift very large animals off the ground and move them around on a track inside the zoo's

veterinary center. These animals can weigh up to 2,000 pounds, and that kind of weight puts a lot of stress on any machine.

Right now, the lift is not stable enough. It shakes or sways too much when it's in use. This is dangerous for the animal and the people who work with them. It also makes it harder for the animal to stay calm and safe seeing as they're only partially sedated. The current lift lacks support from side to side (left to right) and from front to back.

Our main goal is to redesign the lift to improve stability. This means we want it to hold still and steady while lifting and moving an animal. We are going to fix this by using stronger materials and changing how the parts connect.

Why the Problem Matters:

This is not just a school project. The work we are doing can actually help real animals and people. That is why the Nashville Zoo is involved. Their veterinarians need to perform medical procedures on big animals. These animals can't walk into a room and lay on a table very easily like humans can.

Below is a look at the surgical room and table where animals are operated on.

They need help to move safely. Without a working lift, the veterinarians and animal caretakers have to find other ways to move the animals. This can be stressful for the animals and dangerous for the zoo staff. Dr. Schwartz and Mr. Wancea are counting on us to fix this issue so their team can work more safely and effectively.

The stakeholders include:

- The Nashville Zoo as a whole

- Dr. Heather Schwartz (Director of Animal Health)
- Stephen Wancea (Hospital Manager)
- Zoo veterinarians and vet techs
- The animals themselves

Understanding the Current Design:

We are not starting from scratch. The teams from the past two years have already built and tested different prototypes. Their designs got better over time, but there is still work to be done.

Two years ago, the team added something called an “X beam” to help stabilize the frame. This beam was shaped like an X and was supposed to keep the whole lift from shaking. It didn’t work very well, though. So, last year’s team removed the X beam and tried using a single straight beam down the middle instead. That design was better, but it still had problems.

The main issue was that only the two beams in the middle were stable. The ones on the outside were wobbly, and that made the whole structure unreliable. So now it’s our turn to take the best parts from past designs and improve on the weak points.

Our Materials:

Thankfully, the previous teams left us some materials. These include:

- Six metal beams
- Lots of bolts and connectors
- About 30 feet of green ratchet strap
- An engine leveler
- Ratchets for securing straps
- Other small tools and parts

These materials give us a good starting point. We will try to use as much of this as we can to save time and money. If we need more supplies, we might visit the store or check with the school's robotics shed.

Measurements and Requirements:

To build the right kind of lift, we need to know the exact measurements of the space and equipment at the zoo. Some of the important measurements we got are:

- The I-beam (which the lift will roll on) is 3.25 inches wide.
- It has a 1 7/16 inch throat and a ½ inch anchor point.
- There is 16 inches of clearance from the wall.
- The bumpers are 15 inches apart.
- There is 3 inches of vertical clearance from the I-beam to the top of the hoist.
- There are 8 inches from the bottom of the I-beam to the roof.
- The tarp (which supports the animal) is 6 feet 5 inches long and 4 feet 7 inches wide.

All of these numbers matter. They tell us how big or small our design can be. If our lift is too big, it won't fit. If it's too small, it won't be strong enough. We also have to make sure the lift can move on the track system without getting stuck.

Design Goals and Challenges:

Our goal is to build a prototype of the lift that is much more stable than the current one. One idea we have is to use engine levelers – these are tools used in car

repair shops to lift heavy engines. They are very stable and strong, so they might work well in our design.

We are also going to use a stronger cross beam to connect everything more tightly. If we can keep the whole frame from moving side to side or front to back, we'll have solved the main problem.

One big challenge is that we don't have all the information yet. After our next visit to the zoo, we'll know more and be ready to start building. But even then, we might run into problems while prototyping. We may find that some parts don't fit or that our design isn't as stable as we thought. If that happens, we'll go back, make changes, and try again.

Measurable Design Requirements

Based on the zoo's needs and the measurements we collected, our design must meet the following requirements:

1. Must lift up to 2,000 pounds.

2. Must fit within the I-beam structure (3.25 inches wide).

3. Must not interfere with the 8 inches of roof clearance.

4. Must use or fit with the 6'5" x 4'7" tarp.

5. Must be stable from left to right and front to back.

6. Must move smoothly on the track system.

These are measurable goals. We can test them and see if our lift meets each one. If it doesn't, we'll know exactly what to fix.

Conclusion:

This project is more than just a school assignment. It's a real opportunity to solve a real-world problem. The Nashville Zoo needs our help, and we are determined to give them a working solution.

We are taking what we've learned from the previous teams, combining it with our ideas, and building something better. Our main focus is fixing the stability issue. With help from our mentors and guidance from the zoo, we believe we can make a strong, safe, and useful animal lift.

This experience is teaching us about engineering, teamwork, problem-solving, and the real-world impact of what students can do. We are proud to be part of Team Breliconi, and we're excited to continue building and improving this project.